SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Total Marks: 50

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I A. Tharun kumar Reddy (192572068) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Tharun

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- *Library pane showing multiple VMs = multiple instances/regions.*
- *VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.*
- *Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.*
- *VM → Manage → Clone = replica/auto-scaling demonstration.*

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem solving Task

(1) Compare Type-1 and Type-2 Hypervisors:-

Type-1:-

Runs directly on physical hardware,

Example:- VMware, ESXi, Microsoft of Hyper-V.

Type-2:-

Runs on top of a host operating system.

Example:- VMare, Workstation Pro, oracle VirtualBox.

feature	Type-1	Type-2
Performance	High	Lower due to host OS overhead.
Security	Better isolation	Depends on host OS security.
Management	Suitable for large data	Easy for personal use and testing.
Reliability	High	Depends on host OS stability.
Usage	Cloud Servers	Development and learning.

Conclusion:-

Type-1 hypervisors are the better choice for Mission-Critical cloud applications because they Provide high Performance.

(2) Virtualization Architecture and Workload Isolation:-

Virtualization allows Multiple virtual Machines (VMs) to run on the same Physical Server while Maintaining Separation between workloads.

Finance:-

uses isolated VMs to Protect financial transactions and Confidential data.

Healthcare:-

uses Separate VMs to Protect Patient records and Medical information.

Student Portals:-

The uses separate VMs to Manage Student details and academic Service.

-> The hypervisor allocates CPU, Memory, Storage and Network resource to each VM. Access Control and Virtual Networks help Prevent a unauthorized access between workloads.

Two Major attack Surfaces:-

1. Hypervisor Vulnerabilities.
2. Virtual Network attacks.

Conclusion:-

Strong VM isolation, regular Security updates, firewalls, and access, controls, cloud Servity.

(3) VM Host utilization Analysis:-

Given host utilization data.

Parameters	Value	Problem
CPU utilization	92%	CPU overload
Memory utilization	88%	High Memory usage.
Storage I/O latency	35 MS	Slow Storage operations.
Average VM boot time	170 seconds	Slow VM startup.

Likely bottlenecks:-

- CPU
- Memory
- Storage
- Boot time.

Corrective actions:-

1. Add CPU resources (or) distribute VMs across Multiple hosts.
2. Increases avlable Memory and reduce unnecessary Memory usage.

Conclusions:-

CPU, Memory, and Storage improvements can be reduce bottlenecks and improve VM Performance.

tware - Defined Data Center (SDDC):-

A software - Defined Datacenter uses software manage and automate computing, storage and working resources.

enefits compared with traditional Datacentre.

omponents	SDDC Advantages.
Compute Virtualization	Creates and Manages VMs quickly
Storage Virtualization	Combines storage resources into a flexible Storage Pool.
Network Virtualization	Creates virtual Networks and applies Security Policies through software.
Automation	Reduces Manual work and configuration errors
Scalability	Resource can be expanded when demand increases.

Example:-

In a traditional datacenter, administrators may need to configure physical servers and switches manually.

Conclusion:-

SDDC improves agility, Scalability, resource utilization, and Managment.

(5) Multi - cloud Federation:-

Multi-cloud federation allows users to access service from different cloud providers using a trusted identify system.

Problems:-

Identity MisMatches occur when cloud providers use different usernames, identity formats, authentication Method.

Proposed Secure design:-

1. use a Central identity Provider (IdP).
2. Implement federation protocols.
3. use single sign-on.
4. Apply role-based access.
5. Protect Privacy.
6. Enable Multi-factor authentication
7. Monitor Access.

Conclusion:-

Identity federation, Standardized Protocols, least-Privallage access, and Privac Protection help -Establish Secure the Communication between cloud Providers.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded :